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Evaluation of the drought resilience of indigenous irrigation water systems: a case study of dry zone Sri Lanka

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E-mail: shinichiro@civil.nagoya-u.ac.jp**Keywords:** irrigation system performance, water duty, paradigm shift, socio-hydrology, human-agricultural systems, yoda ela, Sri Lanka

Abstract

The wave of modernization and globalization in the last century has rapidly involved a technological paradigm shift from indigenous irrigation water systems to modern systems in arid regions. Despite interest in the drought resilience of indigenous water systems, the impact of this paradigm shift on drought resilience remains poorly understood because previous studies have focused on fixed irrigation water systems. To fill this gap, we investigated the drought resilience of an indigenous and modern irrigation water system that coexists in the drought-prone Mahaweli H region of the Sri Lankan dry zone. To explain drought resilience, we quantified the historical irrigation system performance (1985–2021) of both water systems using the water duty indicator (i.e., the volume of water required to cultivate a unit land area). The statistical Pettitt test was used to detect significant change points in the time series of water duty, and we divided the time line into few periods based on the change points. Furthermore, a quantitative trend analysis of several socio-hydrological variables and a qualitative analysis of their socio-hydrological backgrounds with triggers of water duty were conducted to explain drought resilience path dependency in modern and indigenous water systems. The results indicated a higher drought resilience is embedded in the indigenous system as the mean water duty increment in drought years compared to non-drought years is 16.4% for the indigenous system and 58.3% for the modern system. In addition, drought resilience pathways that elucidated by water duty change points also demonstrated that indigenous water system features a higher drought resilience compared to the modern water system. The findings of this comparative study can contribute to the design of drought resilience improvement strategies in arid region irrigation water systems in a more comprehensive manner.

1. Introduction

Early societies in arid regions invented indigenous irrigation water systems as declining precipitation intensified aridification and threatened human survival. These indigenous human-agricultural systems are highly case based on the spatial, temporal, and socio-economic conditions that originated from them (Laureano 2007). From the ancient times, these systems have evolved resiliently along many generations to support complex civilizations (Laureano 2007, Dharmasena 2010, Koç 2018, Angelakis *et al* 2020, Jana and Tamang 2023, Srivastava and Chinnasamy 2023).

Previous studies have captured the intrinsic resilience attributes related to indigenous irrigation systems in arid lands. These are not merely isolated technical solutions that have emerged to solve the specific issue of water shortage; rather, they display multifunctional benefits to human and natural systems (Panabokke *et al* 2002, Laureano 2007, Dharmasena 2010, Manuel *et al* 2018, Oyonarte *et al* 2022, Jana and Tamang 2023, Weerahewa *et al* 2023). Furthermore, complex interactions between landscapes and communities in indigenous human-agricultural systems have exhibited socio-hydrological resilience by highlighting community values and social cohesion (Gunda *et al* 2018), shared resources and mutualism embedded within the community structure

(Turner *et al* 2016), local knowledge about water management, the ability to adapt to new water management practices, and the hydrological resilience of irrigation systems (Fernald *et al* 2007, 2012, 2015).

However, socioeconomic and political trends in the last century have resulted in the rapid expansion of modern irrigation water systems, including dam reservoirs, canals, and diversion structures in arid regions for food production combined with peasant settlements (Müller and Hettige 1995). This has resulted in a paradigm shift in irrigation water systems from indigenous to modern, with changes in technological norms. Nevertheless, despite the increasing interest in drought resilience, previous studies have assumed a fixed type of irrigation water system (Fernald *et al* 2015, McNamara *et al* 2021, Oyonarte *et al* 2022), and the impact of this paradigm shift, which has occurred in many parts of the world (Tempelhoff 2008, Aubriot 2022), has not yet been considered. Understanding the impact of irrigation water system transformation on resilience attributes can support strategies to improve said resilience in human-agricultural systems in arid regions in a more comprehensive manner (Mustafa and Qazi 2007, Leibundgut and Kohn 2014). However, knowledge of dynamic drought resilience is crucial for comprehending the coevolution of humans and drought systems in dry zone hinterlands. In arid lands, this coevolution is a highly concerning phenomenon in the field of socio-hydrology (Sivapalan *et al* 2012, Han *et al* 2017, Medeiros and Sivapalan 2020), which aims to understand the interactions and coevolution of coupled human-agricultural systems (Van Loon *et al* 2016, Wens *et al* 2019, Streefkerk *et al* 2023).

How drought resilience is affected by transformation of irrigation water system from indigenous to modern types? To tackle the question, we explored how coexisting modern and indigenous water transfer systems respond to perturbations comparatively, targeting in the dry zone irrigation case study analysis in Sri Lanka. In this study, we aimed to evaluate the dry season irrigation system performance and its temporal variations in both modern and indigenous water transfer systems, which offer a good proxy for the state of the human-agricultural system. In addition, a quantitative trend analysis of several socio-hydrological variables and a qualitative analysis of their socio-hydrological backgrounds with triggers of irrigation performance were conducted to explain drought resilience path dependency in modern and indigenous water systems. Finally, we discuss contrasting drought resilience trends in modern and indigenous water systems that emerge through the interaction of coupled human-agricultural systems in the Sri Lankan dry zone.

2. Methodology

2.1. Study area

The dry zone of Sri Lanka, the target site of this study, contains the remains of indigenous irrigation systems (Imbulana *et al* 2006, Murphey 1957). Historical records suggest that the dry-zone water management system was initiated in the 4th century BC, abandoned in the middle of the 13th century AD, and then modified during the British colonial government in the 19th century AD (Abeywardana *et al* 2018). The rapid expansion of large- and medium-scale irrigation developments to colonize the dry zone for food production after independence in 1948 transformed irrigation water systems from indigenous to modern.

The ancient engineering feat Yoda Ela (YE), built by King Dhatusena (459AD) was an inter-basin water transfer canal in ancient times and located in the Sri Lankan dry zone (Brohier 1937). This national resort has endured for more than 1500 years, making it a unique single-bank waterway in the dry zone until it was abandoned in some reaches and partially integrated into a modern water resources development project referred to as the Mahaweli Development Project (MDP) in the 1970s (Fernando 1980). Today, YE and a modern irrigation canal named New Jaya Ganga (NJG), which was built under the MDP, coexist under the Kalawewa reservoir and administered by the Mahaweli H administration (figure 1(b) and table 1). Meanwhile, the low precipitation recorded during the dry season (May to September) has increased the drought risk at the target site. According to previous drought reports (DIMS 2009, ACAPS 2019, MPADMREA 2019), the area was exposed to major drought progressions in 1989, 1992, 2001, 2004, 2012, 2014, and 2019. Thus, the drought resilience of human-agricultural systems plays a major role in achieving sustainability. The coexistence of indigenous and modern irrigation systems provides a valuable opportunity to evaluate the impact of a paradigm shift on drought resilience.

2.2. Data analysis

Hydrological ecosystem services offer a good proxy for human–water interactions and measure the state of socio-hydrological systems (Mao *et al* 2017). In that sense, irrigation system performance parameters are effective in describing the drought resilience of human-agricultural systems. Here, we calculated irrigation system performance in terms of water duty and cropping intensity in dry cultivation seasons and then compared modern and indigenous systems. The duty of water is the relation between a given quantity of water and the area which it serves and it can be expressed as the ratio between irrigated area and water supplied (Abernethy 1990,

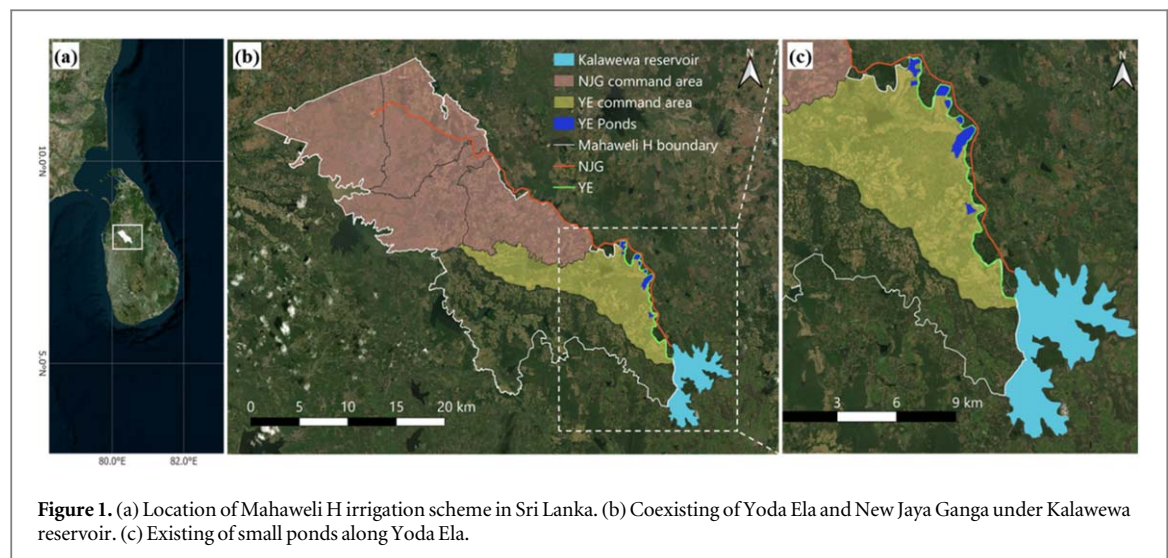


Table 1. Characteristics of Yoda Ela (YE) and New Jaya Ganga (NJG).

Canal name	Command area (ha)	Population as 2020 (Resident Project Manager's Office, System H 2020)	Length (km)
YE (indigenous)	4721	23,698	28
NJG (modern)	14017	115,046	46

Santhi and Pundarikanthan (2000). It is used to set fundamental guidelines for water usage in irrigation contexts (Wescoat 2013), especially in water deficient environments (Abernethy 1990). In previous studies, Chemjong and Wijesekera (2017) adopted water duty to evaluate actual irrigation water utilizations by comparing recommended and actual water duties. Santhi and Pundarikanthan (2000) measured the water distribution equity in distributary irrigation canals and Aheeyar *et al* (2007) described the irrigation water supply performance of a system by using water duty. Meanwhile, cropping intensity is defined as the ratio between the actually cultivated and the total arable areas (Bouwman 1997). Not only in measuring the extent of agricultural land use in local irrigation systems (Aheeyar *et al* 2007, Zwart and Leclert 2010, Dembele *et al* 2012), cropping intensity is also used in many global agricultural land use evaluations (Zhang *et al* 2021, Liu *et al* 2021).

Next, we detect the significant change points in the time series of water duty from 1985 to 2021 using the change-point detection method and divide the time line into few periods based on the change points. We then performed a quantitative trend analysis on several socio-hydrological variables and a qualitative analysis of their socio-hydrological background with triggers of water duty change points using policy documents and governmental reports. To achieve this, we obtained dry season water supply in million cubic meters (mcm), agricultural extent (ha), and seasonal rainfall data from the Water Management Secretariat (WMS) of the Mahaweli Authority of Sri Lanka (MASL). The necessary socioeconomic data were obtained from the MASL annual reports, Central Bank of Sri Lanka, and Department of Census Sri Lanka.

2.2.1. Estimating irrigation system performance

We used a time series of water duty and cropping intensity to evaluate irrigation performance, as defined in equation (1) (Santhi and Pundarikanthan 2000) and equation (2) (Bouwman 1997). When calculating the water duty of each irrigation scheme for the comparison, we excluded climatic and geological effects causing water transfer because both systems are located nearby. In addition, governance attributes are assumed to have no significant impact on the comparison, as both canal systems are managed by the same Mahaweli H administration. However, it is important to account for the variability of canal lengths as canal lengths of YE and NJG are substantially different. As conveyance efficiency (e_c) mainly depends on the length of canal and the type of soil (Brouwer *et al* 1985), we estimated field water supply by using e_c and avoided the length effect in the water duty comparison. According to the technical guidelines of the Mahaweli H administration, the present water-supply scheduling system assumes an e_c of 70% for the YE system and 60% for the NJG system (Resident Project Manager's Office, System H 2021). In particular, field water duty was derived for YE and NJG as in equation (1).

$$\text{Water duty}(m) = \frac{\text{Canal water supply (mcm)} \times e_c}{\text{Agricultural extent (ha)}} \quad (1)$$

$$\text{Cropping intensity (\%)} = \frac{\text{Agricultural extent (ha)}}{\text{Command area (ha)}} \times 100 \quad (2)$$

where canal water supply is the total volume of water supplied to the irrigation canal from the reservoir during the dry season per year and agricultural extent denotes the total area of cultivated land (including paddy and field crops) during the same season and year. The command area, which is a fixed value, indicates the total maximum area of land that can be cultivated under each canal.

The time-series water duty of YE and NJG were compared after smoothing the fluctuations. Similarly, the cropping intensity was smoothed before the comparison. The smoothing process was conducted using Brown's simple exponential smoothing model, which uses an exponentially weighted moving average model (Brown and Meyer 1961).

2.2.2. Detecting change points

We deployed the rank-based nonparametric Pettitt's statistical test to detect time-series change points (Pettitt 1979). This test has been widely applied to determine sudden shifts in hydrological time series (Ahmad and Deni 2013, Dehling *et al* 2013, Guerreiro *et al* 2014, Malekian *et al* 2019, Fooladi *et al* 2021, Li *et al* 2022). Considering the random variables $x_1, x_2, \dots, x_T$ with a change point at x_{t_0} , they are mathematically divided into two segments at t_0 with distinct distribution functions: $F_1(x)$, $t = 1, 2, \dots, t_0$ and $F_2(x)$, $t = t_0+1, \dots, T$. It tests null hypotheses, $H_0: F_1(x) = F_2(x)$ where no change point exists and alternative hypothesis, $H_a: F_1(x) \neq F_2(x)$ where a change point exists at t_0 . To estimate the change-point, the statistical parameters were defined as follows:

$$U_{t,T} = \sum_{i=1}^t \sum_{j=t+1}^T \text{sgn}(x_i - x_j), \quad 1 \leq t < T \quad (3)$$

Where,

$$\text{sgn}(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases} \quad (4)$$

The most probable change point t_0 will be found when,

$$K_{t_0} = U_{t_0,T} = \max |U_{t,T}|, \quad 1 \leq t < T \quad (5)$$

The significant probability related to K_{t_0} is approximated as:

$$p \approx 2 \exp \left(\frac{-6K_{t_0}^2}{T^3 + T^2} \right) \quad (6)$$

Given a certain significance level α , if $p < \alpha$, H_0 is rejected and we conclude that t_0 is a significant change point with α level of significance. In detecting the water duty change points in this study, we use $\alpha = 0.05$ as the threshold level of p value. This indicates that the validity of the statistically significant change-point decision was at 95%. Once a change point was detected, the series was divided into two segments, and Pettitt's test was conducted for each segment until all significant change points were detected.

3. Results

3.1. Irrigation system performance gap between water transfer systems

We estimated the irrigation performance indicators, water duty and cropping intensity as shown in figures 2(a) and (b). According to figure 2(a), up to the early 1990s, both systems showed noticeable increases in the water duty trend values. Both then depicted a gradual decline until 2011. After 2011, the NJG water duty showed a gradual increase and a stable trend up to 2021. In contrast, the YE has shown a gradual decrease in values over the last decade. Regarding cropping intensity, after the 1990s, the YE system showed a higher cropping intensity than NJG (figure 2(b)). Both NJG and YE increased their cropping intensity from 2000 to 2011. In the last decade, when the NJG cropping intensity showed a declining trend, the YE showed a stable trend.

In general, the water duty is inversely proportional to rainfall. When an area receives low rainfall during the season, the amount of water supplied through the canal increases. Thus, the water duty increases. Because of this, the water duty has risen significantly (figure 2(a)) in most of the major drought years, given the system. Importantly, in these drought years, a significant gap was observed in the water duty values between NJG and YE, where YE maintained a significantly lower value than NJG. The mean water duty increment in drought years compared to non-drought years is 16.4% for YE and 58.3% NJG. Overall, YE, the indigenous system, exhibited low water duty and high cropping intensity trends during the dry seasons.

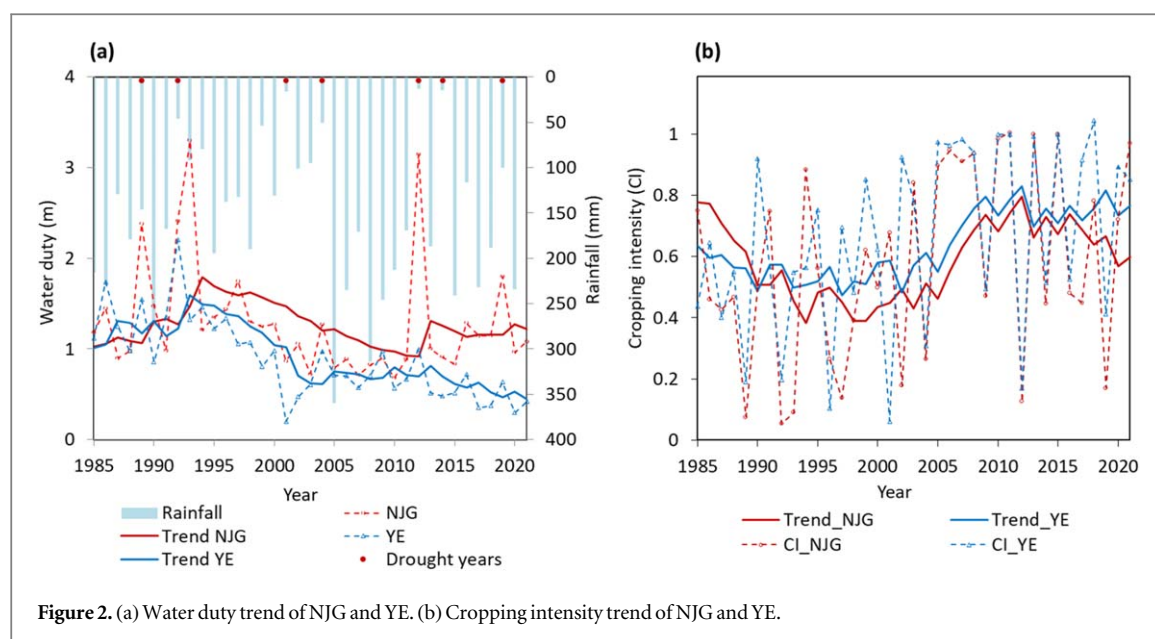


Figure 2. (a) Water duty trend of NJG and YE. (b) Cropping intensity trend of NJG and YE.

3.2. Significant change points of water duty

We applied the Pettitt statistical method to detect significant change points in water duty and divided the timeline into different periods. The first change point in the NJG was detected in 2000, while the second change point was observed in 2011 (figure 3(a)). The average water duty of 1.51 m during 1985–2000 (Period 1) was significantly reduced to 0.87 m in the term 2001 to 2011 (Period 2) in NJG. Later, from 2012 to 2021 (Period 3), the average water duty of NJG significantly increased to 1.33 m. However, the change points of water duty in YE were limited to a single point that occurred in 2000 (figure 3(b)). The average water duty, 1.25 m in Period 1 has dropped significantly to 0.59 m in the following period in YE (figure 3(b)).

Figures 3(c)–(h) depict the quantitative trend analysis of the socio-hydrological variables. As the temporal water duty demonstrated a large variation between the indigenous and modern irrigation systems, in the following section, we discuss the socio-hydrological background and triggers that influenced these heterogeneous behaviors in each period (table 2).

3.2.1. Historical qualitative analysis

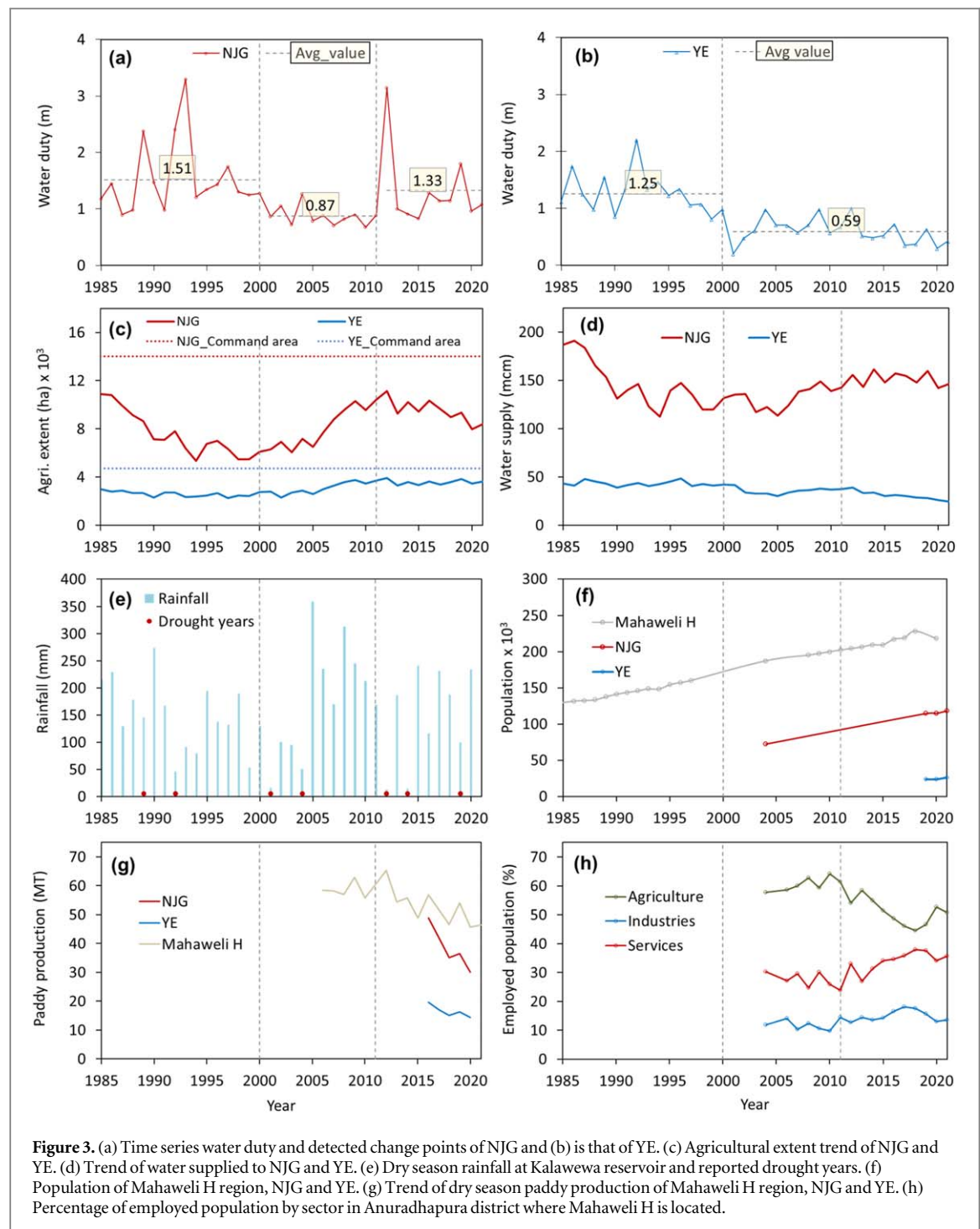
Period 1: 1985–2000, Business as usual

The trend of agricultural extent showed a gradual decline in the NJG (figure 3(c)). Meanwhile, the water supply trend also demonstrated a declining path but reported fluctuations in the NJG (figure 3(d)). Because trends were calculated based on moving average values, this can be attributed to the fact that trend patterns are the result of reporting two major drought events in 1989 and 1992. However, despite the drought events, YE showed a stable trend in agricultural extent (figure 3(c)) and water supply (figure 3(d)). The reported high water duty in major drought years could increase the average water duty in the period compared to the following periods in both NJG and YE. Although participatory irrigation management (PIM) was adopted in 1988 (Samad and Vermillion 1999), it has not shown significant improvements in irrigation performance levels, as farmers tend to consider irrigation systems as something outside their own responsibility (Aheeyar 1997, Samad and Vermillion 1999).

During 1998–2003, the bulk water allocation (BWA) policy and the PIM were both implemented in the Mahaweli H region (Panapitiya *et al* 2008). It is argued that promising pre-agreed water quantities to farmers and fostering ownership of the system by allowing the operation and maintenance of irrigation infrastructure under the BWA has changed farmers' perceptions of water use and irrigation management. Consequently, this has reduced crop failure incidents and increased the agricultural extent (Aheeyar *et al* 2007). Therefore, we interpret that the breakpoint in 2000 in both the NJG and YE systems was a result of the implementation of BWA in the Mahaweli H region.

Period 2: 2001–2011, Increase in farming

After adopting the BWA in 2000, the agricultural extent significantly increased in NJG and gradually increased in YE (figure 3(c)). The extended fertilizer subsidy in 2005, which set fertilizer prices at a very low and fixed value, regardless of the world market price (Weerahewa *et al* 2010) could promote more cultivation in the dry season. In addition, significant rainfall reported from 2005–2011 (figure 3(e)) can influence cultivation in the dry season.



This is also reflected in information related to the employed population by sector. In the district where the Mahaweli H irrigation scheme is located, the population employed in agriculture has shown an increasing trend (figure 3(h)). However, when the water supply trend in NJG followed a slightly increasing path (figure 3(d)), YE followed a constant path (figure 3(d)). The water conservation ability of the YE may have affected this.

Although two major droughts were reported in 2001 and 2004, water duty has not significantly increased in specific years of both systems relative to Period 1, except for YE in 2001 (water duty of YE dropped (figure 3(b)) because water supply through the canal was stopped due to a management decision). This may be due to a change in behavioral norms influenced by BWA. Nevertheless, as the agricultural extent increased significantly compared to the water supply in both NJG and YE, the average water duty could be maintained at a low value, compared to Period 1.

Period 3: 2012–2021, Away from farming

Figure 3(h) shows that the percentage of the agriculturally employed population is decreasing, while industrial and service employment is increasing in the district in Period 3. This can be linked to the effects of the

Table 2. Socio-economic characteristics, drought occurrences and agricultural management by periods.

Period	Period 1 (1985–2000)	Period 2 (2001–2011)	Period 3 (2012–2021)
Socio-economic	Introduced participatory irrigation management in Sri Lanka Introduce bulk water allocation policy in Mahaweli H region	Extended fertilizer subsidy program launched in Sri Lanka End of civil war in the country	Diversified employment opportunities opened in industry and services sectors
Major droughts	1989 and 1992	2001 and 2004	2012, 2014 and 2019
Agricultural management	Farmers tend to see irrigation system as an outside of their own responsibility	Fostered ownership of the irrigation system and changed farmer perception in water use and irrigation management in both NJG and YE	Reduced agricultural extent in NJG due to combined effect of socio-economic dynamics and drought occurrences

end of the civil war in the country in 2009, when non-agricultural sectors opened up in a diversified economy in the country (World Bank 2016). District socioeconomic trends can greatly impact dry season agriculture in the Mahaweli H region and, consequently, NJG, as it shares 55% of the area of the Mahaweli H region. Therefore, the decreasing percentage of the agriculturally employed population could be linked to farmers' tendency to avoid cultivation during the dry season in the NJG.

Thus, a significant declining trend in the agricultural extent (figure 3(c)) was observed, and as a result, the trend of paddy production also demonstrated a declining trend (figure 3(g)) in NJG. Moreover, in this case, three major drought events in 2012, 2014, and 2019 contributed to the agricultural extent trend drop in the NJG, along with regional socioeconomic dynamics developed after ending the civil war in the country. Thus, linking these phenomena to the 2011 water duty break point. In NJG, the average water duty increased compared to Period 2, as the agricultural extent dropped and the water supply was stable (figure 3(d)). Conversely, the agricultural extent trend in the YE was constant (figure 3(c)), and the production drop was marginal (figure 3(g)). Moreover, in the YE, the water supply trend showed a slightly decreasing trend (figure 3(d)). Taken together, the YE followed a trend similar to that in Period 2, demonstrating low a water duty.

4. Discussion and conclusion

This study examined the impact on drought resilience due to paradigm shift of irrigation water system from indigenous to modern. We conducted a case study by deploying indigenous (YE) and modern (NKG) irrigation water systems that coexist in the drought-sensitive Mahaweli H region in the Sri Lankan dry zone. We argue that the concept of water duty, which bridges the water and agricultural extent, is a promising tool for analyzing the drought resilience state of human-agricultural systems, as it is important to use indicators that depict multi-directional interactions.

Water duty analysis revealed two phenomena in this study. First, YE demonstrated low water duty compared to NJG in the dry seasons, particularly in drought years. Secondly, in NJG, based on two significant change points detected, time line could be divided in to three different periods: 'Period 1:1985–2000, business as usual', 'Period 2:2001–2011, increase in farming' and 'Period 3:2012–2021, away from farming'. However, the YE followed a contrasting trend, as it did not detect a second change point and transitioned to Period 3. The transition from Period 1 to Period 2 occurred in both systems because of the implementation of the BWA policy, while the transition from Period 2 to Period 3 in the NJG could triggered by the end of the civil war in the country and the occurrence of droughts at the site.

The historical transition of water systems has led to the change of technological norms to cater to socioeconomic demands. The indigenous YE has been identified as a single-bank water transfer canal (see figure 1(c)), which has ultimately evolved into a series of ponds along the channel, demonstrating an effective method for harvesting rainwater and natural overland flow (Jayasena *et al* 2021). Overall YE has maintained 0.42 mkm^{-1} slope and NJG has a slope of 0.603 mkm^{-1} (Rathnayake and Jayasena 2020). These features demonstrate that the YE can be affiliated with an elongated reservoir that allows water resources to be as static as possible by storing them in online irrigation tanks (Panapitiya *et al* 2008). In contrast, NJG is aimed at providing high discharge of irrigation water faster to paddy lands in downstream. This has resulted a deep and mostly straight main canal which can be viewed as a dynamic water source (Panapitiya *et al* 2008) that depends highly on reservoir compared to YE. Moreover, we observed that shallow, temporary storage and release of water significantly recharge the groundwater table, make the surrounding land marshy, and support riparian habitats in YE compared to NJG. Indigenous systems that still remain in use in arid regions of other countries

demonstrate similar conjunctive use of ground and surface water. Over the generations, the so-called *acequias* systems in United states (Fernald *et al* 2015) and Spain (Martos-Rosillo *et al* 2019, Oyonarte *et al* 2022, Civantos *et al* 2023) have demonstrated hydrological connectivity between surface and shallow groundwater and proven their efficiency and resilience in dry periods. These studies prove the similarity in irrigation performance outcomes of YE and irrigation water distribution systems in other regions. Therefore, when a drought occurs, required water (from reservoir) to cultivate a unit area in the YE scheme is low as the harvested and stored water in ponds can be used as a buffer. In addition, conjunctive use of ground and surface water in YE also benefited in dry periods compared to NJG. As resilience is conceptualized as a set of systematic absorptive, adaptive and transformative capacities (UNISDR 2009, Mao *et al* 2017), we attributed that series of ponds along the canal increase YE system's drought absorptive capacity.

The emerging water duty patterns in each period show how changes in resilience pathways can be identified over space and time. The qualitative shift in the pathway trajectory (path dependency) led to a weaker or stronger resilience of the irrigation system in each period. As established path dependencies significantly change with detected breakpoints, this could be attributed to the linked adaptive capacity (Wilson 2014) of human-agricultural systems. Wilson (2014) further argues that positive transitional ruptures are always linked with anthropogenic factors and adaptive capacity in this context is more closely associated to 'adaptive human potential.' This confirms that both YE and NJG improved their resilience states after adopting the BWA policy in 2000, which influenced changes in human agricultural behavior. Exiting the established pathway and changing direction toward stronger or weaker resilience expresses to which degree the community is prepared to face change. In addition, the social memory of historical decision-making is key to the direction of path dependency (Davoudi 2012). Thus, we could argue that locked-in in the established positive pathway in Period 2 indicates a stronger adaptive capacity of the YE compared with the NJG. It could also raise that continuing the established resilience pathway is a result of embedded higher absorptive capacity of YE compared to NJG. Nevertheless, due to lack of disaggregated social systems data, it creates a space for future research.

The main difference between YE and NJG is the physical water environment. According to Fischer and Sanderson (2022), the physical water context shapes the cultural decision-making processes of farmers. They found stronger water conservation norms among farmers in wetter contexts, and weaker norms among farmers in more arid contexts. From a biological viewpoint, this can be interpreted as maladaptation, as when there is less water, farmers are less motivated to save it. Nevertheless, humans are also social creatures as they respond to social, political, economic, and cultural aspects (Fischer and Sanderson 2022). In our case study analysis, the YE with water-retaining pond structures corresponded to a wetter water context. The NJG receives water only when the reservoir releases water, which corresponds to an arid environment. In this sense, we can argue that farmers in YE have stronger water conservation norms than those in the NJG. This might be delineated by the low water duty in the dry seasons in YE compared to that in NJG. Although cultural characteristics are pivotal in drought adaptation, it is crucial to identify individual motivations such as risk perceptions, attitudes, and abilities, in addition to cultural aspects (Keshavarz and Karami 2016). Therefore, prior to making aggregated conclusions, an analysis is required to distinguish between farmers in indigenous and modern water systems.

In conclusion, we evaluated how the paradigm shift of the irrigation water system from indigenous to modern in the Sri Lankan dry zone impacted the drought resilience of human agricultural systems through the interaction between technology and society. In recent decades, the Sri Lankan government and non-governmental organizations have focused on improving indigenous irrigation systems (which have been neglected over the years) to enhance the resilience of farming communities in the country's drier zones (World Bank 2023, Green Climate Fund 2019). The emergence of this new paradigm, in which the importance of indigenous systems in drought resilience in arid regions has emerged not only in Sri Lanka, but also in many other parts of the world (Hussain *et al* 2008, Gujja *et al* 2009, Adaptation Fund 2016). However, drought resilience studies have tended to focus on fixed types of irrigation systems, and the impacts of these transformations have mostly been neglected. In this sense, a comparative drought resilience assessment, including coexisting indigenous and modern irrigation water systems, as illustrated in this study, could contribute to the comprehensive design of resilience-improving strategies for human agricultural systems in arid regions.

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Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

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